

PROBLEMS SUBMIT CODE MY SUBMISSIONS STATUS STANDINGS CUSTOM INVOCATION

E. The Beast's Encoded Grid

time limit per test: 1 second

memory limit per test: 256 megabytes

The Beast Adhoom, in its never-ending quest to confound, had left behind a large, mysterious grid plastered across a decaying wall in the Unknown. This grid was filled with strange, whispering letters, a silent message from the dark entity.

The three knew that deciphering these messages was crucial for their escape. They also found a smaller, equally cryptic string of characters. It seemed Adhoom had a unique way of hiding clues: a square area within the large grid would hold the next piece of their puzzle, but only if it contained each specific letter from the smaller string at least a certain number of times. The required count for each letter c was exactly $F(c)$, which was the number of times c appeared in the smaller string.

Their task was to find the smallest possible square area in the grid that met these conditions. If no such square existed anywhere in the vast grid, then Adhoom's riddle would remain unsolved, and they would be truly lost.

Help the three find the minimum possible area of such a square.

Input

The first line of the input contains n, m, k ($1 \leq n, m \leq 1000, 1 \leq k \leq n \cdot m$).

The next n lines consist of a string of length m , describing the grid of lowercase letters.

The next line contains string s of length k .

All strings consist of lowercase letters.

Output

Print a single integer. The minimum possible area of a square that contains each letter c at least $F(c)$ times, where $F(c)$ is the frequency of c in s .

If such square doesn't exist, print -1.

Examples

input	Copy
5 5 5 zzzzz dazzz hdzzz amsss zzzzz adham	
output	Copy
9	

input	Copy
3 3 5 sam eah hh1 sameh	
output	Copy
9	

Game of Coders 4 - Over the Garden Wall

Finished

Practice



→ About Contest



[Contest website](#)

→ Virtual participation

Virtual contest is a way to take part in past contest, as close as possible to participation on time. It is supported only ICPC mode for virtual contests. If you've seen these problems, a virtual contest is not for you - solve these problems in the archive. If you just want to solve some problem from a contest, a virtual contest is not for you - solve this problem in the archive. Never use someone else's code, read the tutorials or communicate with other person during a virtual contest.

Start virtual contest

→ Clone Contest to Mashup

You can clone this contest to a mashup.

Clone Contest

→ Submit?

Language: GNU G++23 14.2 (64 bit, ms)

Choose file: Choose File No file chosen

Submit

→ Last submissions

Submission	Time	Verdict
331021798	Jul/27/2025 01:30	Happy New Year!

→ Contest materials

- Statements (pdf) (en)